

Everett Richards

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Overview

I am a dedicated computer scientist who is passionate about using technology to make transportation systems more safe, efficient, and equitable. I am currently working towards my PhD in Computer Science at the University of Delaware, working in the CAR Lab with Prof. Weisong Shi. I am seeking an industry research role in applied intelligent transportation and/or robotics.

Education

University of Delaware

Newark, DE

PhD Computer Science (Advisor: Weisong Shi)

Aug 2026–Present

- Working on connected and autonomous vehicle research.

San Diego State University

San Diego, CA

BS Computer Science & BS Applied Mathematics, 4.0 GPA

Aug 2023–May 2026

- *Summa Cum Laude*
- Most Outstanding Computer Science Graduate
- Most Outstanding Mathematics & Statistics Graduate
- ACM Chapter President

Research Experience

NSF Research Fellow – Worcester Polytechnic Institute

May–Aug 2025

Advisor: Ziming Zhang (Vision, Intelligence, and System Lab)

Worcester, MA

- Conducted robustness evaluations through targeted injection of Gaussian noise into 3D LiDAR point clouds, simulating sensor degradation due to low-resolution sensors and adverse weather conditions.
- Proposed and empirically validated a novel noise-aware training curriculum, achieving up to a 40% improvement in model robustness under challenging real-world scenarios.

NSF Research Fellow – UC San Diego

Sept 2024–June 2025

Advisor: Hao Su (SU Computer Vision Lab)

La Jolla, CA

- Conducted experiments on the impact of Gaussian noise injection in robotic imitation learning models.
- Achieved statistically significant R^2 values between 0.91 and 0.99, supporting my argument that Gaussian noise induces a sigmoidal performance decay curve.

NSF Research Fellow – University of Delaware

June–Aug 2024

Advisor: Lena Mashayekhy (Intelligent Edge Systems Lab)

Newark, DE

- Developed and validated two algorithms to improve object detection accuracy in autonomous vehicles.
- Achieved up to 70% accuracy improvement over legacy methods, validated on a four-robot testbed.
- Presented paper at the IEEE World Congress on Services 2025 in Helsinki, Finland.

Publications

- **E. Richards**. “Task-Aligned Stability Analysis of Vision-Language Models for Autonomous Driving Hazard Detection.” 2026 International Conference on Machine Learning (ICML) Workshop on Combining Theory and Benchmarks. Seoul, South Korea, 2026.
- **E. Richards** and L. Dai. “Modeling Imitation Learning Robustness to Noisy Demonstrations via Sigmoid Degradation.” 2025 IEEE MIT Undergraduate Research Technology Conference (URTC). Cambridge, MA, 2025.
- **E. Richards**, A. Lopez, J. Morales, and Z. Zhang. “From Chaos to Clarify: Strengthening 3D Collaborative Autonomous Vehicle Perception with Noise-Aware Training.” 2025 IEEE MIT Undergraduate Research Technology Conference (URTC). Cambridge, MA, 2025.
- **E. Richards**, B. Thapa, and L. Mashayekhy. “Edge-Enabled Collaborative Object Detection for Real-Time Multi-Vehicle Perception.” 2025 IEEE International Conference on Edge Computing and Communications (EDGE). Helsinki, Finland, 2025, pp. 13-22, doi: 10.1109/EDGE67623.2025.00011.

Instructional Experience

Teaching Assistant, Computer Science
University of Delaware

Aug 2026–Present
Newark, DE

- Teaching Assistant for the Department of Computer and Information Sciences.

Coding Camp Instructor
iD Tech Camps / UC San Diego

Jun–Aug 2026
La Jolla, CA

- Instructed students aged 13-17 on computer programming and video game development.

Tutor, Mathematics & Computer Science
SDSU Math and Science Learning Center (MSLC)

Aug 2024–May 2025
San Diego, CA

- Tutored students in computer science (Java, Python, data structures, algorithms), mathematics (calculus, discrete math, real analysis), and physics.
- Participated in professional development workshops to enhance tutoring skills.
- Certified Level 1 Tutor by the College Reading and Learning Association (CRLA).

Instructional Assistant, Discrete Mathematics
SDSU Dept. of Mathematics and Statistics – Instructor: Vadim Ponomarenko

Jan 2024–May 2025
San Diego, CA

- Led weekly office hours for ten groups of 6 to 8 students, covering proof techniques, recursion, and boolean algebra.
- Developed and discussed weekly focus exercises to reinforce key topics.

Leadership Experience

Vice Chair, CTRL SDSU
Coalition of Tech Representatives and Leadership (CTRL) at SDSU

May 2025–Present
San Diego, CA

- Served on a board of leaders from various Computer Science student organizations at SDSU.
- Represented CTRL at CS faculty and Industrial Advisory Board (IAB) meetings.
- Co-Directed the biannual student-led "Innovate4SDSU" Hackathon.
- Oversaw the NXP Engineering Bootcamp, in which 15 students worked with NXP industry mentors over the course of the semester to create AI-driven solutions to chip design problems.

University Council Representative
Associated Students of San Diego State University

May 2025–Present
San Diego, CA

- Represent 8,000 College of Sciences students on the University Council.
- Oversee financial appropriations for a non-profit with over \$45,000,000 in annual revenue.
- Drafted a successful resolution advocating for increased transparency in changes to mandatory student fees.

President, ACM SDSU
Association for Computing Machinery (ACM) Student Chapter at SDSU

Apr 2024–Present
San Diego, CA

- Led SDSU's chapter of the world's largest professional organization dedicated to computing.
- Secured \$2,500 sponsorship from Google to fund events and initiatives.
- Presented 10+ workshops on topics like machine learning, web development, and version control.

Awards and Honors

Academic Awards:

- Most Outstanding Computer Science Graduate, SDSU Dept. of Computer Science (2026)
- Most Outstanding Mathematics Graduate, SDSU Dept. of Mathematics and Statistics (2026)
- Cert. of Excellence in Advanced Programming Languages, World Computing Organization (2025)
- Dean's List, SDSU College of Sciences (2023-2026)

Scholarships:

- Steven Michael Rogers Scholarship in Mathematics (2025)
- George A. Hansen Scholarship (2025)
- Deloitte Foundation Scholarship (2024)
- Mensa Foundation Scholarship (2024)
- Raytheon Systems Scholarship (2024)
- Intuit STEM Scholarship (2023)

Skills and Strengths

Technical Skills:

- Deep learning with TensorFlow, Keras, PyTorch, OpenCV
- Data analysis with NumPy, Pandas, Matplotlib, and Scikit-learn
- Computer vision, including CNNs & deep learning
- Computer networking and edge computing
- Theoretical and applied mathematics

Professional Skills:

- Research, literature review, and technical writing
- Public speaking and presentations
- Leadership and organizational management
- Tutoring and mentoring
- Drafting LaTeX documents
- Publishing papers in reputable journals